

Coughing Up Blood

(Hemoptysis)

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Coughing up blood from the respiratory tract is called hemoptysis. The amount of blood produced can vary from a few streaks of blood mixed with normal sputum to large amounts of pure blood. Other symptoms, such as fever and difficulty breathing, may be present, depending on the cause of hemoptysis.

Causes of Coughing Up Blood

Although hemoptysis can be frightening, most causes turn out not to be serious. Blood-streaked sputum is common in many minor respiratory illnesses, such as upper respiratory infections (URIs) and viral bronchitis. Sometimes the cause is blood from the nose that has traveled down the throat and then is coughed up. Such blood is not considered hemoptysis.

Common causes

Infection is the most common cause (see table [Some Causes and Features of Hemoptysis](#)). In adults, many cases are caused by:

- [Acute respiratory infections including bronchitis](#) and pneumonia
- [Bronchiectasis](#), which is an abnormal, irreversible widening of part of the breathing tubes or airways (called bronchi)
- [Tuberculosis](#)
- Lung cancer

In children, common causes are:

- A lower respiratory tract infection
- A foreign object that has been inhaled (aspirated)

Less common causes

However, cancer that has spread to the lungs from elsewhere in the body rarely causes hemoptysis. Fungal infection with *Aspergillus* (called [aspergillosis](#)) is recognized as a cause but is not as common as cancer.

Other causes include a blood clot in an artery in a lung ([pulmonary embolism](#)) and, less commonly, inflammation of blood vessels ([vasculitis](#)) in the lung, such as [anti-glomerular basement membrane disease](#) (also called Goodpasture syndrome) or [granulomatosis with polyangiitis](#). [Heart failure](#) and [heart valve disorders](#) can rarely cause mild hemoptysis.

Massive hemoptysis

Massive hemoptysis is coughing up so much blood as to be life threatening within 24 hours. The most common causes include the following:

- [Lung cancer](#)
- [Bronchiectasis](#)
- Some pneumonias, including [tuberculosis](#)

Risk factors

Some conditions increase the risk that hemoptysis is caused by a serious disorder:

- [Human immunodeficiency virus \(HIV\) infection](#) (for Kaposi sarcoma, tuberculosis, and fungal infections)
- Use of medications that suppress the immune system called immunosuppressants (for tuberculosis and fungal infections)
- Exposure to tuberculosis
- A long history of smoking (for cancer)
- Recent bed rest, surgery, or long-distance travel (for [pulmonary embolism](#)), cancer, and weight loss

Evaluation of Coughing Up Blood

The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with hemoptysis, the following symptoms are of particular concern:

- Large amounts of blood coughed up
- Shortness of breath
- Signs of significant blood loss (weakness, dizziness when standing up, thirst, sweating, and a rapid heart rate)
- Weakness or fatigue
- Back pain
- Having a tracheostomy

When to see a doctor

People with warning signs should go to the hospital immediately. People without warning signs who have risk factors for serious disorders and those with more than just blood-streaked sputum should see a doctor in a day or two.

If people have only blood-streaked sputum (which is usually caused by an upper respiratory infection), a doctor's evaluation is not as urgent. People can call a doctor, who can decide whether and how rapidly they need to be seen based on their symptoms, medical history, and other factors. Typically, a short delay is not likely to be harmful.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history and then do a physical examination. What doctors find during the history and physical examination often suggests a cause and the tests that may need to be done (see table [Some Causes and Features of Hemoptysis](#)).

Doctors ask:

- When the person started coughing up blood
- How long the coughing has been going on
- Whether anything specific triggers it (such as cold, exertion, or lying down)
- About how much blood is coughed up (such as streaks, a teaspoonful, a cupful, a few milliliters)
- Whether the person has other symptoms, such as fever, weight loss, chest pain, or leg pain

Doctors determine whether blood was actually coughed (and not vomited or dripped down the back of the throat from a nosebleed).

Doctors ask people about their medical history (if not already known) and their risk factors for causes. A history of frequent nosebleeds, easy bruising, or liver disease suggests a possible [blood clotting disorder](#). Doctors review the medications the person is taking to check for those that inhibit clotting (anticoagulants).

During the physical examination, doctors review vital signs to check for fever, rapid heart or breathing rate, and test for a low oxygen level in the blood. They do a full heart and lung examination, inspect the neck veins for signs of fullness such as bulging, and check the legs for puffiness ([edema](#)). Puffiness in one leg may indicate a blood clot ([deep vein thrombosis](#)). Puffiness in both legs may indicate [heart failure](#). Doctors also examine the abdomen, skin, and mucous membranes. The person is asked to cough during the examination. If any blood is coughed up, the doctor notes its color and the amount of blood. Doctors also check the nose and mouth for bleeding sites.

Clues from the history and examination help doctors determine the cause. A sensation of postnasal drip or any bleeding from the nose, particularly without coughing, may mean that the blood being coughed up has dripped down the back of the throat from the nose. Nausea and vomiting of black, brown, or coffee-ground-colored material usually means that the blood is from the stomach or intestine and is being vomited and not coughed. Frothy sputum, bright red blood, and, if the amount is massive, a sensation of choking usually mean that the blood is from the trachea or lungs (called true hemoptysis).

If cough is acute and if the person is otherwise in good health and has no risk factors for tuberculosis, fungal infection, or pulmonary embolism, the cause is usually an acute respiratory infection, such as [acute bronchitis](#). If coughing up blood is caused by a heart or lung disorder, the person has almost always already been diagnosed with that heart or lung disorder. That is, coughing up blood is usually not the first symptom of a heart or lung disorder.



Some Causes and Features of Hemoptysis

Cause	Common Features*	Tests†
A blood clotting disorder Use of anticoagulants (as used to treat pulmonary embolism, blood clots in the legs, or atrial fibrillation or to reduce the risk of clots after certain heart procedures) Use of medications that dissolve clots (thrombolytic medications, as used to treat a heart attack or stroke)	Sometimes bleeding from other sites, such as the nose or digestive tract (seen in stool) In people taking anticoagulants or thrombolytic agents	Blood tests that assess the blood's ability to clot
	Sometimes a family history of a blood clotting disorder	
Bronchiectasis	A chronic cough and mucus production in people with a history of recurring infections	High-resolution CT of the chest Sometimes bronchoscopy
Bronchitis • Acute bronchitis • Chronic bronchitis	<i>Acute:</i> A cough that may or may not produce sputum (productive or nonproductive) and sometimes symptoms of an upper respiratory infection (such as a stuffy nose)	<i>Acute:</i> A doctor's examination
	<i>Chronic:</i> A productive cough on most days of the month or for 3 months of the year for 2 successive years in	<i>Chronic:</i> A chest x-ray



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